José Miguel Ramírez Gutiérrez

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Education

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM) Querétaro, Mexico
B.S. in Robotics and Digital Systems Engineering (GPA: 3.7/4.0) *Aug. 2024 – Expected 2028*

- Socioeconomic Scholarship

Experience

Automation Developer – PEPS Validation June 2026
Valeo (Contractor) – Querétaro, Mexico

- Automated manual Key Fob functional validation testing by architecting a complete positioning-system software/hardware solution, saving engineers up to 8 hours of testing daily.
- Designed and validated an autonomous workspace calibration system using 4 HC-SR04 ultrasonic sensors with a halving-approach algorithm, achieving ± 1 cm positioning accuracy over a 390×1300 mm test area; diagnosed and resolved sensor crosstalk through circuit-level analysis.
- Built a Python/Tkinter GUI for real-time lab instrumentation control and automated execution of validation scripts, streamlining hardware communication.

Cloud Integration & IT Specialist 2026 – Present
binarysails – Querétaro, Mexico

- Collaborated to verify and standardize Microsoft OneDrive functionality, remote sharing, and synchronization across multiple external computers.
- Formulated an official instructional curriculum syllabus outlining cloud storage collaboration techniques and synchronization states for workplace environments.
- Developed and delivered training presentations guiding users on effective cloud file management and the transition from local storage to OneDrive structures.
- Professionalized the delivery of project documentation by standardizing the organization of electronic and firmware files within shared cloud drives.

Projects

Custom Linux Single Board Computer (SBC) July 2026
KiCad, Allwinner V3s, Hardware/Circuit Design

- Engineered a custom 4-layer PCB around the Allwinner V3s (ARM Cortex-A7) processor, integrating an EA3036CQB PMIC, Ethernet PHY, and onboard RTC – requiring extensive datasheet interpretation and circuit analysis for component selection and validation.
- Implemented differential-pair routing with controlled impedance to guarantee signal integrity for High-Speed USB 2.0 (480 Mbps) communication.
- Optimized board layout and component placement for automated SMT manufacturing (PCBA), resolving centroid rotation and footprint constraints through systematic schematic and DRC/ERC review.

ESP32 AI Voice Assistant May 2026
MicroPython, KiCad, Embedded Systems

- Designed a custom PCB in KiCad integrating an ESP32 microcontroller, INMP441 I2S digital microphone, PAM8403 amplifier, and SH1106 OLED display; generated and ordered production-ready Gerber files.
- Achieved rapid response times by integrating the Groq API (Whisper + LLaMA 3.1) via a Flask backend deployed on Render.
- Implemented wake-word detection with VAD-based audio capture and power-supply noise reduction.

DetectaGas – IoT Gas Leak Prevention System December 2025
ESP32, Rust, React, MQTT

- Engineered an embedded IoT device using an ESP32 microcontroller and MQ-5 gas sensor circuit to rapidly reduce response time to hazardous LP gas leaks.
- Built a concurrent Rust/Axum backend and React interface, processing continuous sensor telemetry via an MQTT broker.

Academic Microcontroller & Digital Systems Projects

- **8-bit RISC CPU on FPGA** – Designed a custom processor architecture in VHDL on a DE10-Lite board, covering digital logic design and hardware description languages.
- **Pololu 3pi+ Maze-Solving Robot** – Implemented PD control and intersection detection for autonomous navigation.
- **Raspberry Pi Embedded Projects** – Built UART/serial applications (Vigenère cipher tool, Simon Says API with FastAPI/gpiozero) and OLED display scripting.

Technical Skills

- **Languages:** C/C++, Python, Rust, C#, JavaScript, MATLAB, VHDL
- **Embedded & Hardware:** ESP32, ESP8266, Arduino, ATmega328P, KiCad, PCB & Schematic Design, Circuit Analysis, Datasheet Interpretation
- **Communication Protocols:** I2S, SPI, UART, MQTT
- **Software & Tools:** MicroPython, Arduino IDE, Axum, MySQL, Git, React, Flask
- **Spoken Languages:** Spanish (Native), English (Business Fluent)